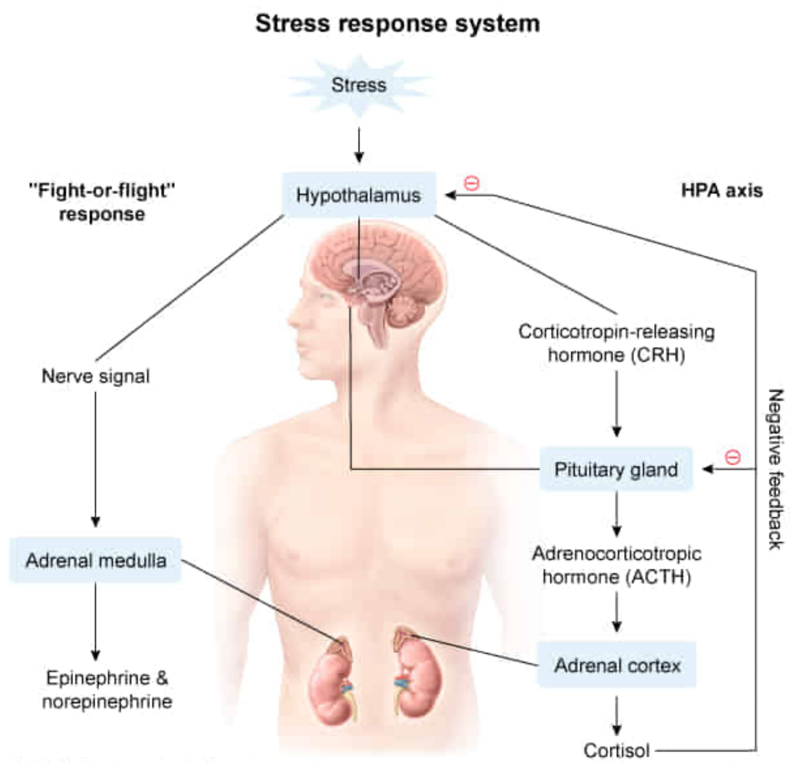
A drug that blocks the release of adrenocorticotropic hormone from the anterior pituitary would most likely:

- A. decrease secretion of cortisol from the adrenal cortex. [90%]
- B. decrease secretion of thyroid-stimulating hormone from the anterior pituitary. [6%]
- C. increase secretion of oxytocin from the posterior pituitary. [1%]
- D. increase secretion of calcitonin from the thyroid. [1%]

Correct
90% answered correctly

Explanation:



HPA is the hypothalamic-pituitary-adrenal axis.

The adrenal glands are located atop each kidney. Each adrenal gland can be subdivided into two portions: a central inner portion called the adrenal medulla, and an outer portion called the

adrenal cortex, which is responsible for secreting corticosteroids (eg, cortisol, aldosterone).

The **hypothalamic-pituitary-adrenal pathway** (HPA axis) controls the secretion of glucocorticoids from the adrenal cortex as follows:

1. The **hypothalamus** secretes **corticotropin-releasing hormone** (CRH) in response to low glucocorticoid levels and increased stress.

2. CRH acts on the anterior pituitary, which causes the release of **adrenocorticotropic hormone** (ACTH).
3. ACTH stimulates the **adrenal cortex** to synthesize and release **cortisol**.

4. Cortisol targets tissues (eg, muscle and liver) to increase the ability to cope with stressors. In addition, cortisol functions as a negative feedback signal by inhibiting the secretion of CRH and ACTH.

A drug that blocks the release of ACTH would also block the stimulatory effect of ACTH on the adrenal cortex, thereby **decreasing the secretion of cortisol**.

(Choice B) The secretion of thyroid-stimulating hormone (TSH) from the anterior pituitary is mediated by the HPA axis. In this pathway, secretion of TSH is controlled by the release of thyrotropin-releasing hormone (TRH) from the hypothalamus, *not* ACTH.

(Choice C) The release of oxytocin from the posterior pituitary is stimulated by pressure against the cervix during childbirth. In response to oxytocin, uterine contractions increase, which further stimulate the release of oxytocin until the baby has been delivered. Oxytocin also targets breast tissue to stimulate milk ejection.

(Choice D) In response to an increase in blood calcium levels, the parafollicular cells of the thyroid secrete calcitonin. The primary purpose of calcitonin is to reduce calcium levels; however, the impact on plasma calcium concentrations has been found to be insignificant.

Educational objective:
Corticosteroids (glucocorticoids and mineralocorticoids) are steroid hormones released by the adrenal cortex. The release of cortisol from the adrenal cortex is mediated by the secretion of adrenocorticotropic hormone from the anterior pituitary.

Time spent:0 sec	**QID**:401385
Subject:	**System:**
Biology	Endocrine and Nervous Systems